ADR 001: Multi-Tab Performance & Resilience Mapping

Status: Accepted / Implemented

Context

Standard RVTools snapshots often fail to capture real-time performance bottlenecks. Migrating VMs to IBM Cloud VPC based solely on assigned resources risks undersizing workloads experiencing CPU contention or over-provisioning "Zombie" VMs.

Decision

Implementation of a correlation engine pulling from five distinct data silos (vInfo, vCPU, vHost, vCluster, vDisk).

Key Logic Implemented:

- **Safety Overrides:** Automatic specification matching for workloads with >5% CPU Ready or >3% Co-stop.
- **N+1 Headroom:** Cluster-wide capacity validation ensuring Day 1 high-availability.
- **Density Guardrails:** 6:1 vCPU:pCore ratio tracking.

PROJECT SPECIFICATIONS

RVTools to IBM Cloud VPC

Performance-Aware Migration

Automated transformation of VMware RVTools exports into production-ready IBM Cloud VPC Terraform code.

Core Intelligence

Infrastructure Resilience

Calculates cluster-wide headroom to maintain enterprise-grade uptime during host failures by analyzing the largest failure points vs. total demand.

Cost Optimization

Identifies "Zombie" assets powered on but showing near-zero utilization, enabling teams to reduce projected cloud spend.

Technical Specifications

- **Processing Engine:** Python / Pandas
- **Interface:** Streamlit
- **Output:** Terraform HCL / CSV Audit Trail